FPL-UDA: FILTERED PSEUDO LABEL-BASED UNSUPERVISED CROSS-MODALITY ADAPTATION FOR VESTIBULAR SCHWANNOMA SEGMENTATION

Jianghao Wu[1], *Ran Gu*[1], *Guiming Dong*[1], *Guotai Wang*[1,2], *Shaoting Zhang*[1,2]

[1] School of Mechanical and Electrical Engineering,
University of Electronic Science and Technology of China, Chengdu, China
[2] Shanghai Artificial Intelligence Laboratory, Shanghai, China

ABSTRACT

Automatic segmentation of Vestibular Schwannoma (VS) from Magnetic Resonance Imaging (MRI) will help patient management and improve clinical workflow. This paper aims to adapt a model trained with annotated ceT1 images to segment VS from hrT2 images, without annotations of the latter. The proposed method is named as Filtered Pseudo Label-based Unsupervised Domain Adaptation (FPL-UDA) and consists of three components: 1) an image translator converting hrT2 images to pseudo ceT1 images, where a two-stage translation strategy is proposed to deal with images with VS in various sizes, 2) a pseudo label generator trained with ceT1 images to provide pseudo labels for the pseudo ceT1 images, where a GAN-based data augmentation method is proposed to deal with the domain gap between them, and 3) a final segmentor trained with hrT2 images and the corresponding pseudo labels, where an uncertainty-based filtering is used to select high-quality pseudo labels to improve the segmentor's robustness. Experimental results with a public VS dataset showed that our method achieved an average Dice of 81.52% for VS segmentation from hrT2 images, which outperformed existing unsupervised cross-modality adaptation methods.

***Index Terms*—** Domain adaption, image translation, uncertainty estimation, vestibular schwannoma, pseudo labels.

1. INTRODUCTION

Vestibular schwannoma (VS) is a benign and usually slow-growing tumor that affects the hearing and balance nerves, and even facial nerves and nearby brain structures if the size is large, with an increasing incidence in recent years [1]. Volume measurement of the VS plays an important role for patient management. Segmentation of VS from high-quality Magnetic Resonance Imaging (MRI) is important for accurate volume measurement and monitoring of the VS. As manual segmentation is time-consuming and depends on the operator's experience, automatic segmentation is highly desirable.

Contrast-enhanced T1-weighted (ceT1) and high-resolution T2-weighted (hrT2) MRI are two common modalities for VS imaging. CeT1 images have a high contrast for VS, making it easier to annotate the target. However, there are increasing concerns on the risk of accumulative use of gadolinium contrast agents. In contrast, hrT2 is preferred due to its safety but it has a lower contrast, leading to more difficulties in annotating VS for training an segmentation model.

To deal with this problem, unsupervised cross-modality adaptation methods are potential solutions by learning from annotated ceT1 images and unannotated hrT2 images to segment VS from hrT2 images. Due to the domain shift between the two modalities, e.g., different contrasts and intensity distributions, it is not practical to direccly apply a model trained with ceT1 images to the hrT2 images. Therefore, transferring the knowledge reliably from ceT1 images to unannotated hrT2 images is crucial for the adaptation task [2].

In the literature, several Unsupervised Domain Adaptation (UDA) methods [3, 4, 5, 6, 7, 8] have been proposed for the cross-modality adaptation task, and they are mainly based on feature alignment and image translation techniques. Feature alignment encourages the segmentation model to learn domain-invariant features through Generative Adversarial Networks (GAN) [3]. However, it usually cannot handle pixel-level domain shift well, so it can hardly achieve good results for dense semantic segmentation [4]. Image translation methods, such as CycleGAN [5], implement pixel-to-pixel image conversion by learning from unpaired images from source and target domains. SIFA [6] combines feature alignment and image translation to obtain better performance. Although they can convert a target domain image into a source domain-like image, there may still exist domain gaps between the generated and real source domain images, and it is hard to ensure semantic consistency after translation, leading to unsatisfactory segmentation of target domain images.

Inspired by the above methods and to overcome their limitations, we propose a novel method named as Filtered Pseudo Labels (FPL)-UDA for the VS segmentation task. Our contributions are three-fold. First, to translate hrT2 images into ceT1 images reliably, we propose a two-stage translation method that can improve the translation quality for VS with various sizes. Secondly, to deal with the domain gap between

Corresponding author: Guotai Wang (guotai.wang@uestc.edu.cn).

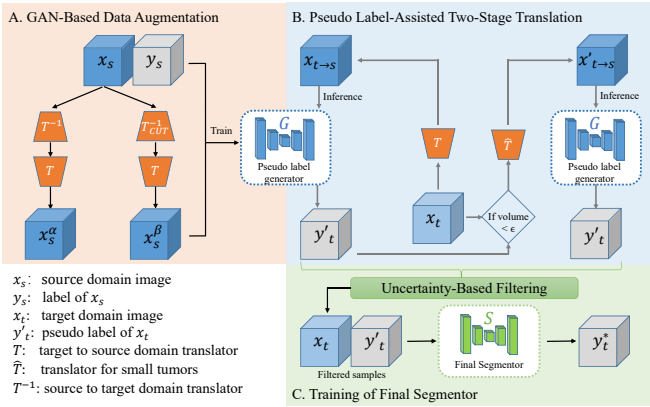

Fig. 1. Overview of our proposed FPL-UDA for VS segmentation. It includes a pseudo label generator, an image translator with two-stage translation and a final segmentor that is trained with target domain images and their pseudo labels.

real ceT1 and pseudo ceT1 images, we propose a GAN-based data augmentation method, which enables a model trained on annotated ceT1 images to generate reliable pseudo label for the pseudo ceT1 images converted from unannotated hrT2 images. Thirdly, we propose an uncertainty-based filtering method to select high-quality pseudo labels for hrT2 images to learn a final segmentation model. The proposed pipeline is validated with a public VS dataset, and the experimental results showed that our method based on pseudo label learning outperformed existing image translation and feature alignment methods for unsupervised cross-modality adaptation.

2. METHODS

Our FPL-UDA framework is illustrated in Fig. 1. It includes a pseudo label generator G, an image translator T and a final segmentor S. Let $\mathbf{X}_s$ and $\mathbf{X}_t$ denote a set of labeled source domain (ceT1) images and a set of unlabeled target domain (hrT2) images respectively, and let x_s and x_t denote an image from $\mathbf{X}_s$ and $\mathbf{X}_t$, respectively, where the label of x_s is y_s. Note that x_s and x_t are unpaired. T converts x_t into a source domain-like image $x_{t\to s} = T(x_t)$, so that the domain gap between $x_{t\to s}$ and x_s is small. Then, the pseduo label generator G trained with $\mathbf{X}_s$ takes $x_{t\to s}$ as input and obtains the pseudo label y'_t for x_t. Finally, x_t and y'_t pairs in the target domain are used to train S for hrT2 images. To improve the performance, we propose GAN-based augmentation for G, pseudo label-assisted two-stage translation for T and uncertainty-based filtering of pseudo labels for S, respectively.

2.1. GAN-Based Data Augmentation for G

We first use $\mathbf{X}_s$ and $\mathbf{X}_t$ to train a CycleGAN [5] for image translation. It consists of a target-to-source domain translator T and source-to-target domain translator T^{-1}. Though $x_{t\to s} = T(x_t)$ is source domain-like, it may still have domain gaps from x_s, which limits the performance of applying G trained with $\mathbf{X}_s$ to $x_{t\to s}$ for obtaining pseudo labels.

To overcome this problem, we use a GAN-based data augmentation strategy to improve the generalizability of G when applied to $x_{t\to s}$. Specifically, we augment x_s by two folds. First, we send x_s to the trained CycleGAN [5] to obtain a pseudo ceT1 image $x_s^\alpha = T(T^{-1}(x_s))$, so that both x_s^α and $x_{t\to s}$ are pseudo ceT1 images, and the domain gap between $x_{t\to s}$ and x_s^α is smaller than that between $x_{t\to s}$ and x_s. Secondly, we use another translator T_{CUT}^{-1} implemented by CUT [9] to translate x_s into a target domain-like image and then convert it to back to a source domain-like image $x_s^\beta = T(T_{CUT}^{-1}(x_s))$. Thus, for each source domain image x_s, it shares the same label y_s with its augmented versions x_s^α and x_s^β. We thus use the set of $\{x_s, x_s^\alpha, x_s^\beta, y_s\}$ for all the source domain images to train a segmentation network G in a fully supervised manner. Then the pseudo label for x_t is obtained by $y'_t = G(x_{t\to s}) = G(T(x_t))$.

2.2. Pseudo Label-Assisted Two-Stage Translation

As the pseudo label y'_t is based on $x_{t\to s}$ given by T, the performance of T has a large impact on the pseudo label quality. In our VS segmentation task, we empirically found that a naive T implemented in CycleGAN [5] translates the background region well but the quality around the VS is not good, especially for small VS, which is mainly due to the fact that CycleGAN [5] is VS-agnositic, and the proportion of small VS is much lower than that of larger VS in practice. Therefore, the discriminator of CycleGAN [5] is more inclined to identify cases with large VS as true, and at the same time, it cannot provide sufficient guidance to synthesizing small VS well due to its bias towards larger VS.

To deal with this problem, we propose a pseudo label-assisted two-stage translation with the following steps. First, based on the pseudo labels, we define a threshold ϵ, and select all the hrT2 images with a pseudo label volume lower than ϵ, leading to a target domain subset $\hat{\mathbf{X}}_t \subset \mathbf{X}_t$. Similarly, for images in $\mathbf{X}_s$, we calculate the volume of VS based on the ground truth label, and select all the images with a volume lower than ϵ, leading to a source domain subset $\hat{\mathbf{X}}_s \subset \mathbf{X}_s$. Secondly, we use $\hat{\mathbf{X}}_s$ and $\hat{\mathbf{X}}_t$ to train another CycleGAN [5] model for image translation with small VS. The target-to-source translator in this new CycleGAN [5] is denoted as $\hat{T}$. Thirdly, for $x_t \in \hat{\mathbf{X}}_t$, its pseudo label is updated as $y'_t = G(\hat{T}(x_t))$.

As $\hat{T}$ is specifically designed for small VS images, it has a better image translation quality than the initial translator T, leading to better pseudo labels for small VS. As shown in Fig. 2, CycleGAN [5] and CUT [9] have good image translation performance for large VS, but their performance for small VS is bad, i.e., they even completely ignore the exis-

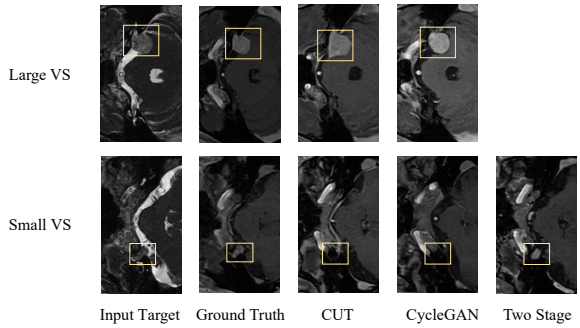

Fig. 2. Visual comparison of image translation obtained by CycleGAN, CUT and two-stage translation. The source and target domains are hrT2 and ceT1 images, respectively. Note that for small VS, the translated images obtained by CUT and CycleGAN have a poor quality around the VS, while the two-stage translation has a better performance.

tence of small VS in the translated images. After our two-stage translation with $\hat{T}$, the small VS is better preserved and enhanced.

2.3. Uncertainty-Based Filtering of Pseudo Labels for S

As the pseudo label y'_t of x_t is generated by G, it is inevitable that y'_t is not always accurate. Training S with inaccurate y'_t will limit its performance. To deal with this problem, we propose an uncertainty-based filtering method to select high-quality pseudo labels to train S for the target domain.

First, we use uncertainty information to estimate the pseudo label quality. Specifically, Monte Carlo Dropout is used for uncertainty estimation due to its simplicity and effectiveness [10]. We use random dropout in the last layers of G, and for $x_{t\to s}$, we conduct K times of inference to obtain K different predicted probability maps after softmax. Let $\bar{P}$ denote the average result across these probability maps. Secondly, for each voxel, we calculate the variance of the foreground probability across the K predictions, leading to a variance map V with the same size as $x_{t\to s}$. We sum the value of each voxel in V to get a naive image-level uncertainty v:

$$v = \sum_i V_i \tag{1}$$

where V_i is the variance of voxel i. Thirdly, as v can be biased towards large VS, we normalize v by the estimated size of uncertain region η. Specifically, we calculate the entropy of voxel i based on $\bar{P}$ and denote it as E_i. η is defined as:

$$\eta = \sum_i \mathcal{H}(E_i - e) \tag{2}$$

where e is a threshold for voxel-level entropy and $\mathcal{H}()$ is the unit step function that takes 0.0 for negative inputs and 1.0 for positive inputs. Then our proposed image-level uncertainty is:

$$u = v/\eta \tag{3}$$

Finally, we rank all the images in $\mathbf{X}_t$ according to the value of u in the ascending order and select the top R images with the lowest image-level uncertainty of the pseudo labels. The selected images in $\mathbf{X}_t$ and the pseudo labels are used to train the final segmentor S in the target domain.

3. EXPERIMENTS AND RESULTS

3.1. Data and Implementation

Our method was evaluated on the public Vestibular Schwannoma segmentation dataset [11], which includes 3D MRI images from 242 patients with VS. Each patient was scanned by ceT1 and hrT2 respectively. We randomly split the 242 patients into 200 for training and 42 for testing. In the training set, ceT1 images with annotations from 100 patients and hrT2 images without annotations from the other 100 patients were used. In the testing set, we only used the hrT2 images for inference. For preprocessing, all the images were cropped by a cubic box determined by the largest possible range of VS in the training set and expanded by a margin, and normalized by intensity mean and standard deviation.

Both G and S were implemented by an existing 2.5D network [1] for VS segmentation from MRI with anisotropic 3D resolution and they were trained with the Dice loss. The networks were implemented in PyTorch, using NVIDIA GeForce RTX 2080Ti GPU. We used the Adam optimizer with momentum of 0.9, initial learning rate of 10^{-3} for training. The total iteration number was 40000, and the batch size was 4. We trained CycleGAN [5] and CUT [9] for 400 epochs. We set ϵ = 2000 voxels, e = 0.2, and R = 40 in the experiment. The segmentation accuracy was measured by Dice score and Average Symmetric Surface Distance (ASSD).

3.2. Comparison with Existing Methods

We compared our method with several existing UDA methods: CycleGAN [5] and Nice-GAN [12] that are based on image translation, and SIFA [6] that is based on feature alignment. They are also compared with the "No DA" lower bound, i.e., directly applying a model trained with $\mathbf{X}_s$ to images in $\mathbf{X}_t$, and with the upper bound, i.e., training the segmentation model in the target domain with full annotations. The quantitative comparison between these methods is shown in Table 1. The "No DA" method obtained an average Dice of 4.02%, indicating a significant domain shift between the ceT1 and hrT2 modalities. The average Dice and ASSD achieved by our FPL-UDA was 81.52% and 2.21 mm, respectively, and the performance is significantly higher than the other four state-of-the-art methods with p-value < 0.05. A visual comparison between these methods is shown in Fig. 3.

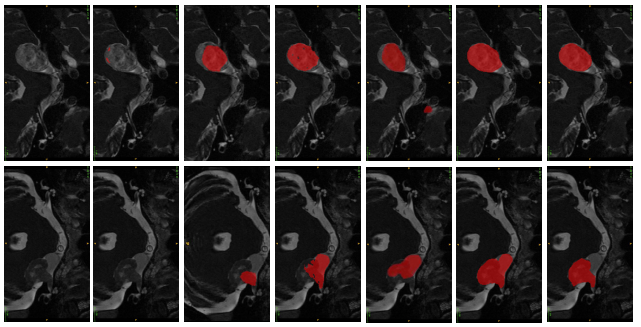

Fig. 3. Visual comparison of different UDA methods for VS segmentation.

Table 1. Quantitative comparison of different UDA methods for VS segmentation in the target domain of hrT2 images.

Method	Dice (%)	ASSD (mm)
No DA	4.02±10.13	32.65±16.87
CycleGAN [5]	57.85±26.97	6.65±11.95
NICE-GAN [12]	41.99±32.68	12.13±19.71
SIFA [6]	50.00±38.22	9.94±14.40
Our FPL-UDA	**81.52±12.01**	**2.21±3.98**
Upper bound	86.92±10.74	1.12±3.20

3.3. Ablation Study

For ablation study, we first investigate the effectiveness of our GAN-based augmentation. The baseline is using CycleGAN [5] and it is compared with Aug1 and Aug2 that mean x_s is augmented by x_s^{α} and (x_s^{α}, x_s^{β}), respectively. Note that all these three methods use T to obtain $x_{t\to s}$ and apply G to $x_{t\to s}$ to get segmentation of testing images. The first section in Table 2 shows that our GAN-based augmentation method can improve the performance of G greatly.

Then, we use pseudo labels obtained by G to train S for the segmentation task, and the results are presented in the second section of Table 2. G-Aug2 + S means we directly used all the pseudo labels to train S, and S-FPL means the uncertainty-based filtering of pseudo labels based on Eq. 3 is additionally used. $+\hat{T}$ means the pseudo label-assisted two stage translation is used. Table 2 shows that both of our FPL and two-stage translation improved the performance of S. We also replaced our FPL by FPL$^{\diamondsuit}$ that means the quality of pseudo labels are measured by Dice between it and the ground truth label, and we selected the R pseudo labels with highest Dice. The last two rows of Table 2 show that our FPL is better than FPL$^{\diamondsuit}$, mainly due to that FPL$^{\diamondsuit}$ is biased towards large VS images that have higher Dice than smaller ones in $\mathbf{X}_t$.

Table 2. Ablation study of our FPL-UDA.

Method	Dice (%)	ASSD (mm)
G-Baseline	57.85±26.79	6.65±11.95
G-Aug1	62.37±25.36	4.22±10.50
G-Aug2	**71.30±17.98**	**2.68±7.28**
G-Aug2 + S	70.14±24.68	3.99±7.91
G-Aug2 + S-FPL	77.58±15.75	3.42±6.38
G-Aug2 + S-FPL + $\hat{T}$	**81.52±12.01**	**2.21±3.98**
G-Aug2 + S-FPL$^{\diamondsuit}$ + $\hat{T}$	80.27±12.93	1.58±4.00

4. DISCUSSION AND CONCLUSION

The two-stage translation in our method is to enhance the image quality for small VS, where the hyper-parameter ϵ was set to an intuitive empirical value based on the characteristics of the VS dataset. On a new dataset, ϵ may also need to be set based on the size distribution of the target lesion or organ. For some applications where the target has a uniform size distribution, the two-stage translation may not be necessary.

In conclusion, we propose a filtered pseudo label-based unsupervised cross-modality adaptation (FPL-UDA) for accurate and efficient segmentation of Vestibular Schwannoma from hrT2 MRI. GAN-based data augmentation is proposed to make the pseudo label generator G obtain high-quality pseudo labels for unannotated target domain images, and pseudo label-assisted two-stage translation is used to obtain better source-domain like images, which also improves the quality of pseudo labels. We also introduce uncertainty-based filtering of pseudo labels for training the final segmentor in the target domain. Comprehensive experimental results on the public VS segmentation dataset validated the effectiveness of our proposed method, and it outperformed several existing unsupervised domain adaptation methods. It is of interest to apply our method to other datasets in the future.

5. COMPLIANCE WITH ETHICAL STANDARDS

This work used the public dataset [11] created from real clinical exams at the Queen Square Radiosurgery Centre. The data was approved by the NHS Health Research Authority and Research Ethics Committee (18/LO/0532) and was conducted in accordance with the 1964 Declaration of Helsinki.

6. ACKNOWLEDGMENT

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